# TESS Hunter: batch-one deep vetting

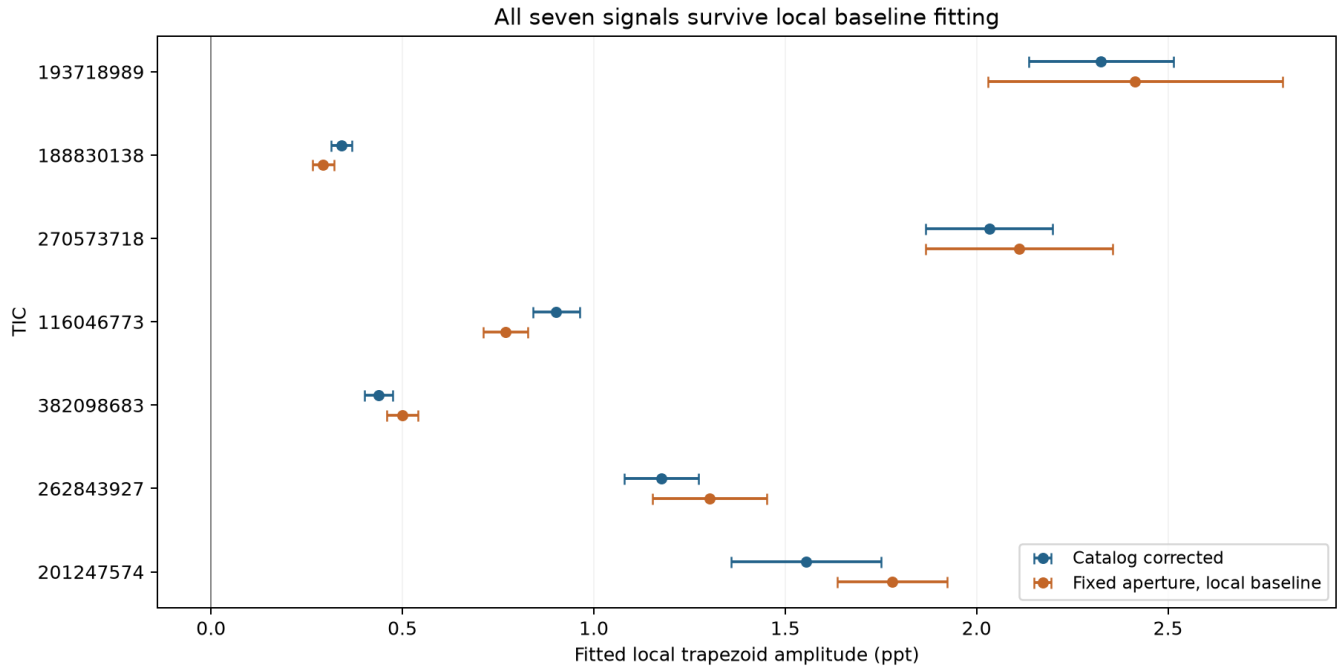
## Seven candidate signals | analysis dated 18 October 2026

**Six signals favor planetary transits; TIC 116046773 remains a serious host-assignment problem.** None is confirmed or validated by this analysis. The strongest source evidence is for TIC 193718989 and TIC 262843927; TIC 201247574 also improves after the aperture audit.

This review remeasured events in the saved corrected and fixed-aperture light curves, downloaded and fitted 25 new TESS image cutouts, checked fresh Gaia fields and current ExoFOP tables, and compared calibrated and empirical source models. It does not simply repeat the pipeline verdicts.

| TIC | Period (d) | Radius* (Earth) | Assessment / localization |
|---|---|---|---|
| 193718989 | 24.532886 | 2.66 | Planet favored<br>Strong target support |
| 188830138 | 1.181024 | 1.43 | Planet favored; host open<br>Weak target preference |
| 270573718 | 41.383701 | 2.12 | Planet favored<br>Target favored; baseline-sensitive sector |
| 116046773 | 5.650272 | 2.29 | Ambiguous blended signal<br>Unresolved; model-dependent |
| 382098683 | 4.351685 | 1.31 | Planet favored; localization modest<br>Modest target preference |
| 262843927 | 24.108929 | 1.99 | Planet favored; upgraded<br>Strong target support |
| 201247574 | 16.560576 | 2.44 | Planet favored; upgraded<br>Good target support |

** Radii are the existing physical-fit estimates assuming the named target is the host. They are not masses and are invalid if a different star hosts the event. TIC 262843927 lacks a reliable formal radius error because the original central-transit covariance is singular.*




Error bars include local residual time correlations and event-to-event scatter, but do not constitute a global search false-alarm calibration. The trapezoid amplitudes differ slightly from box-average and limb-darkened model depths.

# What changed in the evidence

## Signal detection and source identification are separate

All seven primary dips survive local baseline fitting in independently extracted fixed-aperture photometry. This supports real repeating dimming, but a TESS aperture can include many stars. An eclipsing neighbour can produce a shallow planet-like signal after dilution. A small fitted radius only applies after the host is identified.

## Several alarming aperture discrepancies are baseline problems

For TIC 116046773, apparent brightenings in sectors 25, 52 and 53 become positive 0.69, 0.74 and 0.81 ppt dimmings. For TIC 201247574, sectors 95 and 96 become mutually consistent at 1.50 +/- 0.44 and 1.57 +/- 0.46 ppt. TIC 262843927's nominally strong phase-0.3876 dip loses coherent significance under local event fitting. These changes reduce the evidence for false positives, without resolving every source.

## The zero-neighbour summary is unreliable

Every saved summary reports zero neighbours capable of causing the candidate dip, while its detailed entries disagree. A fresh Gaia DR3 cone query finds potential contaminating stars around every target. The report uses explicit positions and flux ratios, not the zero counters. Gaia RP is a proxy for the TESS band, and actual aperture transmission changes the required eclipse depth.

## Saved FPP is not an overall probability that a planet exists

| TIC | FPP (%) | NFPP (%) | Stellar-eclipse scenarios (%) | Planet on companion STP (%) |
|---|---|---|---|---|
| 193718989 | 2.822 | 0.173 | 0.195 | 2.624 |
| 188830138 | 3.845 | 0.765 | 0.913 | 2.905 |
| 270573718 | missing | missing | missing | missing |
| 116046773 | 12.180 | 0.027 | 0.032 | 12.121 |
| 382098683 | 14.940 | 0.084 | 0.103 | 14.819 |
| 262843927 | 4.349 | 0.000 | 0.005 | 4.327 |
| 201247574 | 4.184 | 0.367 | 0.365 | 3.808 |

These are the original batch TRICERATOPS results, not reruns in this review. They are conditional on a transit-like astrophysical signal, the adopted scene, priors and noise treatment. STP is a planet on an unresolved stellar companion and is counted as a false positive for the nominated host. Instrumental false alarms are outside the scenario set. The stellar-eclipse column sums scenario names containing EB. Zero is numerical, not an absolute exclusion. [3]

Published TRICERATOPS validation thresholds commonly used for small candidates are FPP < 1.5% and NFPP < 0.1%, with the physical and observational assumptions checked. None of the saved batch results meets the FPP threshold; TIC 270573718 has no successful value. The public Gaia-constrained result for TIC 193718989 is a different analysis, discussed in its case page. [3,4]

# TIC 193718989

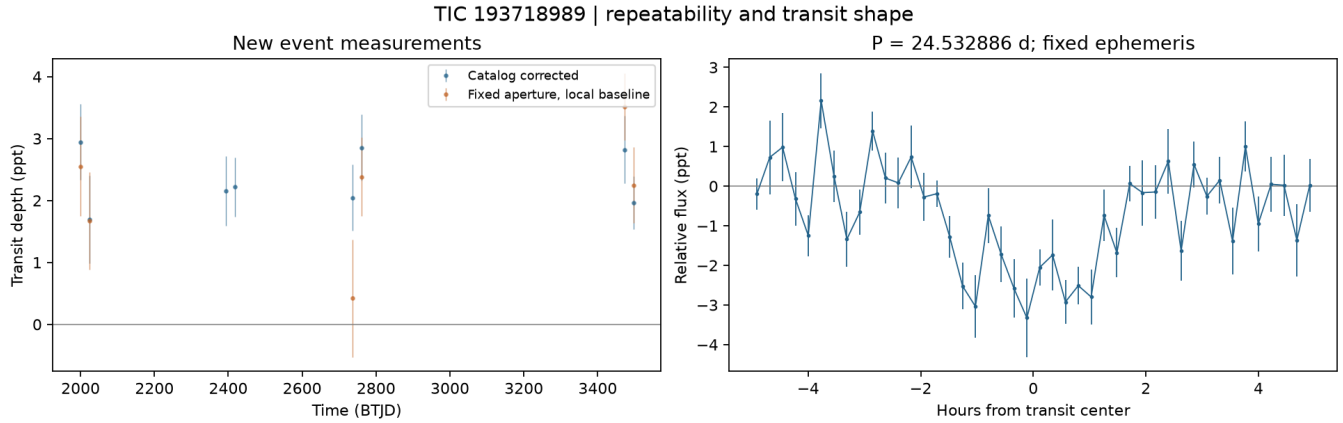
## Strong planet candidate; target strongly favored

The 24.532886-day signal is the best-balanced case in this group. Its repeated narrow dimming, plausible transit geometry, raw-aperture recovery, and localization all favor a planet transiting the nominated star. This remains a candidate, not a confirmed or statistically validated planet.

| Parameter | Value | Evidence |
|---|---|---|
| Period / epoch | 24.53288577 d<br>BTJD 2000.075484 | Existing batch physical fit; BTJD = BJD(TDB) - 2457000 |
| Duration / nominal radius | 3.350 h<br>2.66 +/- 0.29 Earth radii | Conditional on the nominated host |
| New corrected / aperture amplitude | 2.325 +/- 0.189 ppt<br>2.414 +/- 0.386 ppt | 8 / 6 adequately covered events; all signs retained |
| New pixel sectors | 53, 79, 80 | 3 fresh 15 x 15-pixel TESS cutouts |

All eight usable corrected-light-curve events have positive measured depths. A new local trapezoid fit gives 2.325 +/- 0.189 ppt; the saved fixed aperture independently gives 2.414 +/- 0.386 ppt from six adequately sampled events. There is no significant odd/even difference or half-period secondary. The free-fit stellar-density ratio is 0.99, so the planet-like geometry is not merely imposed by the stellar-density prior. The Gaia/VSX rotation flag can explain the slow baseline, but does not by itself explain these narrow, coherent events.

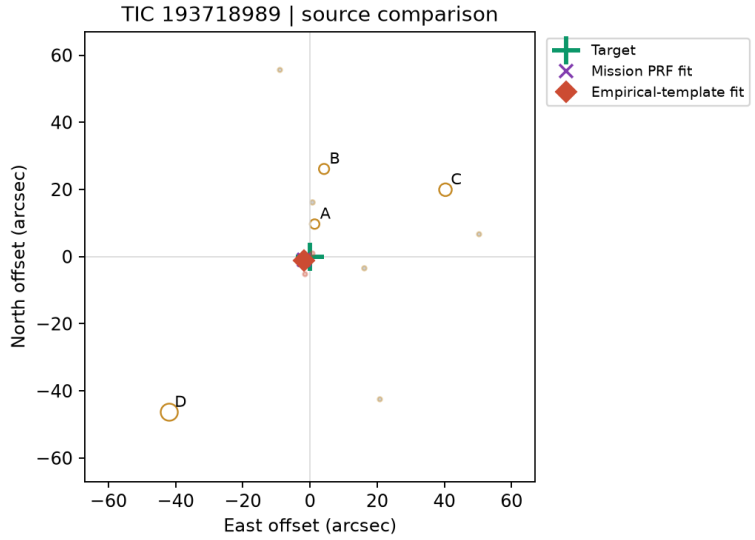
The nearest faint neighbour could mimic the mean observed dip only with an approximately 74% eclipse under an equal-throughput Gaia-RP approximation. A public candidate analysis already reports this same signal at 24.532923 days and 2.58 Earth radii, consistent with the batch fit of 2.66 +/- 0.29 Earth radii. That is an independent analysis of largely the same TESS observations, not new observing confirmation. The saved batch FPP is 2.82%; the public analysis obtains about 2.95% with default priors and about 1.11% with Gaia companion constraints. Those numbers use different assumptions and should not be combined.




**Next decisive step.** Highest priority: resolve the target and the 9.8-arcsec neighbour during a predicted transit, then obtain high-resolution imaging and rerun validation with the measured scene. Attribute the existing public candidate report when discussing novelty.

# TIC 193718989: where is the dimming?

New sector-53, 79 and 80 images all prefer the target among catalogued positions. Joint source fits lie about 2.0 arcsec from the target with the mission PRF and 2.2 arcsec with the empirical template. The closest alternative, Gaia DR3 1346023547768414976, is 9.8 arcsec away at the Gaia epoch; fixing the dimming there worsens the combined diagnostic loss by about 42-44. This is strong relative evidence for the target, but is not a calibrated host probability or proof against an unresolved companion. Leaving out each sector in turn keeps the empirical position within 1.8-3.1 arcsec of the target.




**Nearby alternatives**

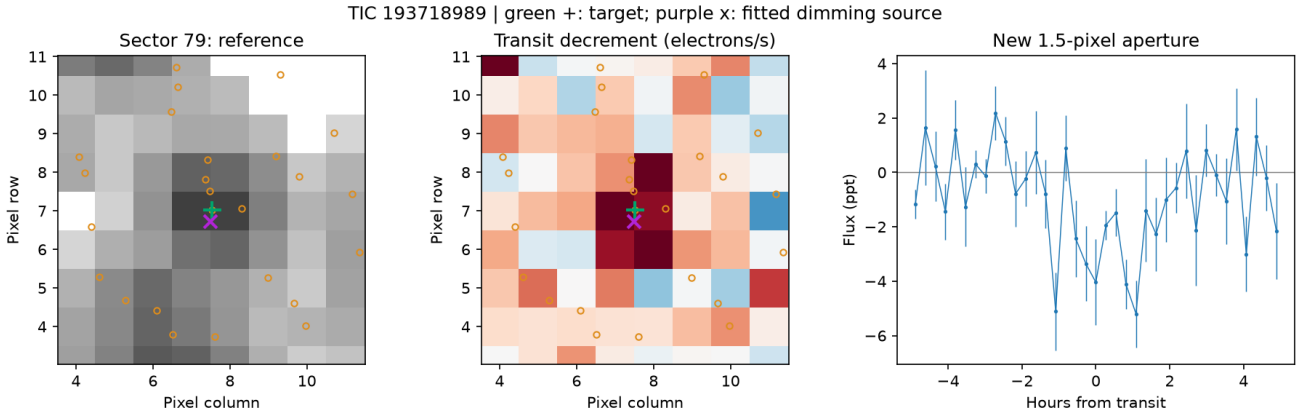
**A**: 9.8 arcsec; eclipse ~74%
Gaia DR3 1346023547768414976

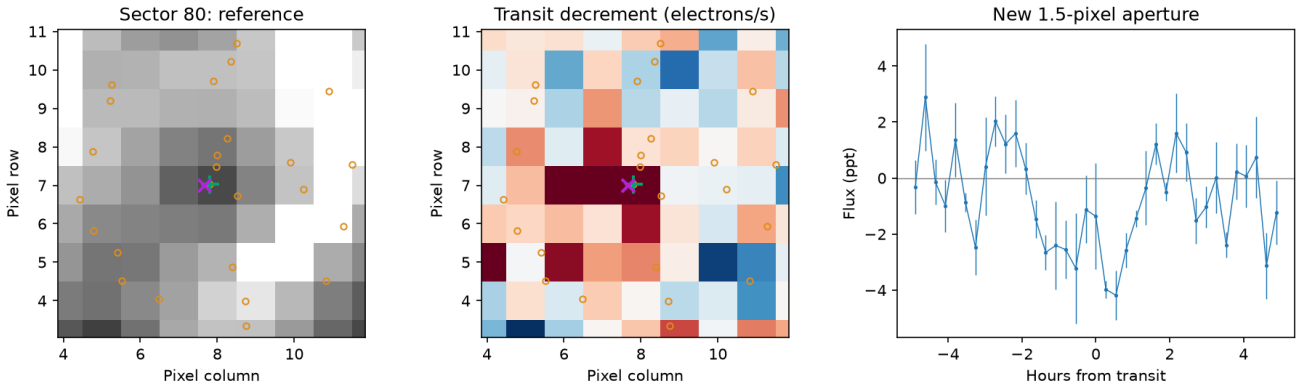
**B**: 26.4 arcsec; eclipse ~95%
Gaia DR3 1346023612193934848

**C**: 45.0 arcsec; eclipse ~18%
Gaia DR3 1346023646553671936

Labels correspond to the sky chart. The full source list is retained in the evidence JSON.

* Gaia-epoch separation, not rounded separation from the older TIC coordinate. ** Mean batch dip, Gaia RP flux ratio and equal target/neighbour throughput; these are illustrative estimates, not strict bounds or exact TESS-band predictions. Faint orange dots are less capable sources; circles are approximately capable. Pale red points show separate-sector empirical source fits. Crosses/diamonds show joint fits.




The illustrated image fits use the linear baseline. Alternative quadratic-baseline results are retained in the data package and discussed where they change the interpretation. No position in this chart is a calibrated exclusion boundary.

# TIC 188830138

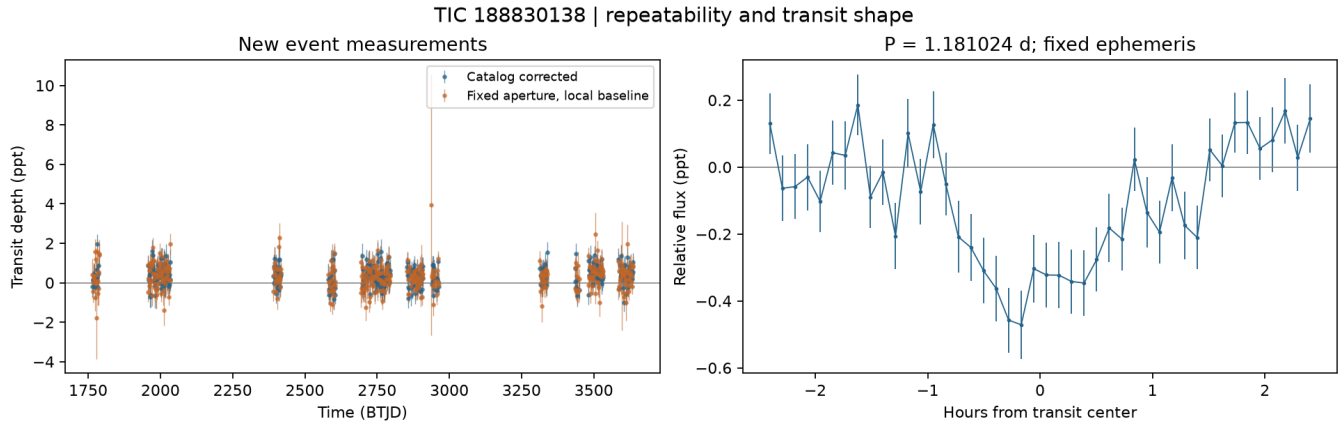
## Strong repeating planet candidate; host not securely localized

The 1.181024-day signal is highly repeatable and compatible with a small planet. Its shallow depth makes source attribution appreciably weaker than the signal detection. I would retain it as a planet candidate while leaving nearby eclipsing-binary contamination open.

| Parameter | Value | Evidence |
|---|---|---|
| Period / epoch | 1.18102404 d<br>BTJD 1764.890889 | Existing batch physical fit; BTJD = BJD(TDB) - 2457000 |
| Duration / nominal radius | 1.640 h<br>1.43 +/- 0.15 Earth radii | Conditional on the nominated host |
| New corrected / aperture amplitude | 0.341 +/- 0.027 ppt<br>0.293 +/- 0.028 ppt | 334 / 390 adequately covered events; all signs retained |
| New pixel sectors | 53, 58, 74, 85 | 4 fresh 15 x 15-pixel TESS cutouts |

The batch retains 304 events across 19 sectors. The new analysis, which uses all adequately covered events rather than only those retained by the pipeline, measures 0.341 +/- 0.027 ppt in corrected photometry and 0.293 +/- 0.028 ppt in the saved fixed aperture. The earlier negative aperture measurements in sectors 17, 51, 58, 60 and 78 become positive or consistent with zero under local baselines. A phase-0.5 secondary is absent. The new odd/even difference is about 2.1 sigma in corrected flux and 1.5 sigma in the aperture, insufficient to identify a doubled-period binary.

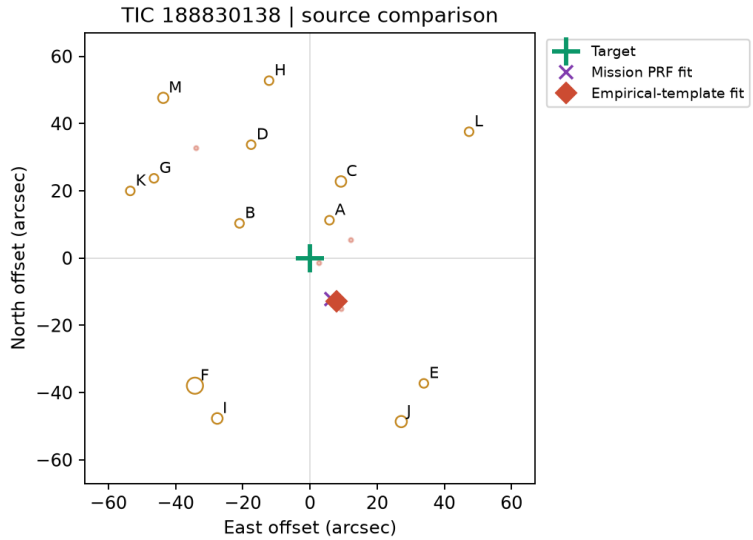
The nearby TIC 188830141, Gaia DR3 1416467677212567424, is 51.1 arcsec away and has RUWE about 10.1. It is the leading nearby-binary alternative in the saved probability calculation. The four-sector image fit disfavors that position relative to the target by about 14 loss units, but this is not a complete exclusion. The closer 12.7-arcsec star could mimic the dip with an approximately 35% eclipse before aperture losses. The 1.43 +/- 0.16 Earth-radius estimate assumes the named host; saved FPP/NFPP are 3.85%/0.765%.




**Next decisive step.** Observe the closest faint neighbours and TIC 188830141 at the 1.181024-day ephemeris. Their eclipses would be much deeper than the 300-ppm target signal. Joint localization using all 19 pixel sectors is a useful next analysis; the four-sector result here remains inconclusive.

# TIC 188830138: where is the dimming?

Four new pixel sectors (53, 58, 74 and 85) provide only a weak preference for the target among catalogued sources. The nearest neighbour, Gaia DR3 1416479423947847296 at 12.7 arcsec, is worse by only about 5 diagnostic loss units. The unrestricted combined source lies 14-15 arcsec from the target and sector positions scatter. A good folded light curve therefore does not justify an on-target claim.




**Nearby alternatives**

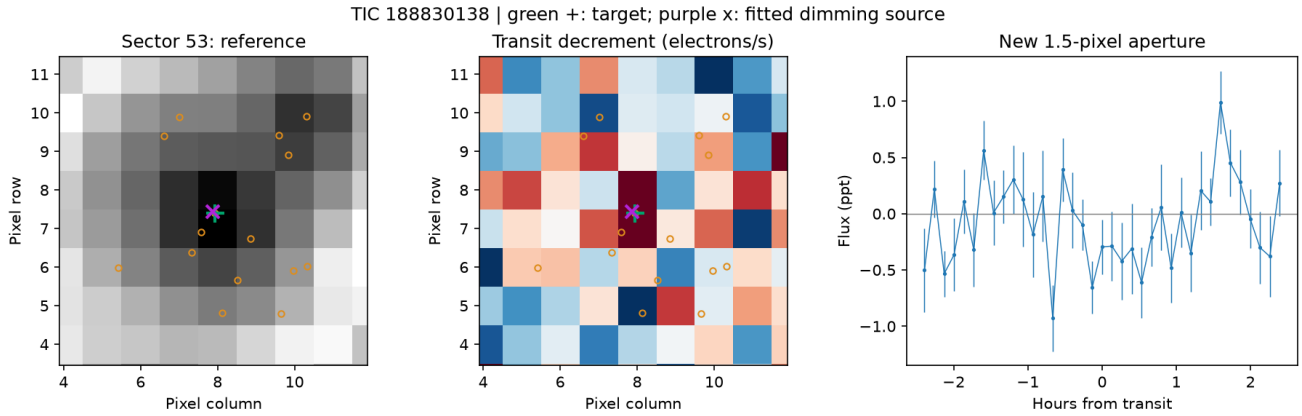
**A**: 12.7 arcsec; eclipse ~35%
Gaia DR3 1416479423947847296

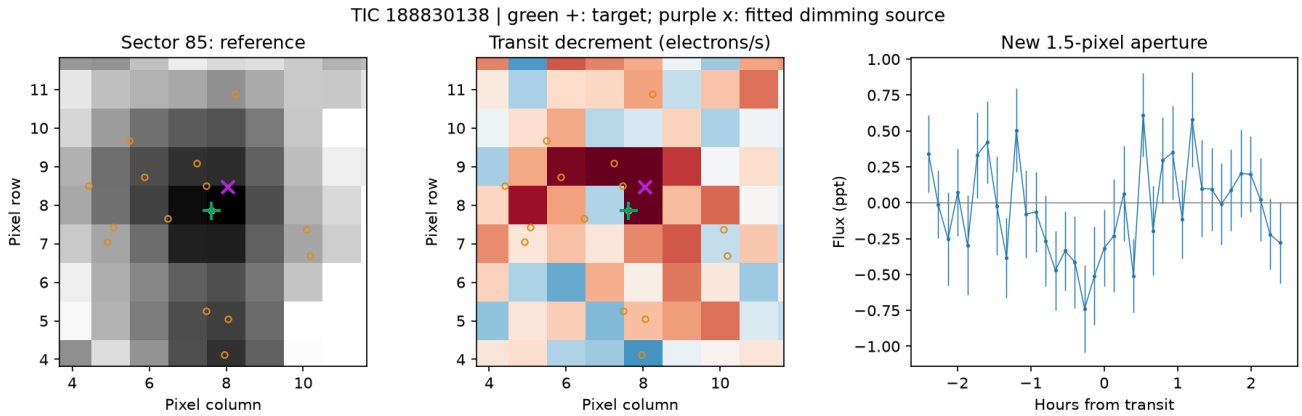
**B**: 23.4 arcsec; eclipse ~38%
Gaia DR3 1416479423947077888

**C**: 24.6 arcsec; eclipse ~10%
Gaia DR3 1416479423947080960

Labels correspond to the sky chart. The full source list is retained in the evidence JSON.

* Gaia-epoch separation, not rounded separation from the older TIC coordinate. ** Mean batch dip, Gaia RP flux ratio and equal target/neighbour throughput; these are illustrative estimates, not strict bounds or exact TESS-band predictions. Faint orange dots are less capable sources; circles are approximately capable. Pale red points show separate-sector empirical source fits. Crosses/diamonds show joint fits.




The illustrated image fits use the linear baseline. Alternative quadratic-baseline results are retained in the data package and discussed where they change the interpretation. No position in this chart is a calibrated exclusion boundary.

# TIC 270573718

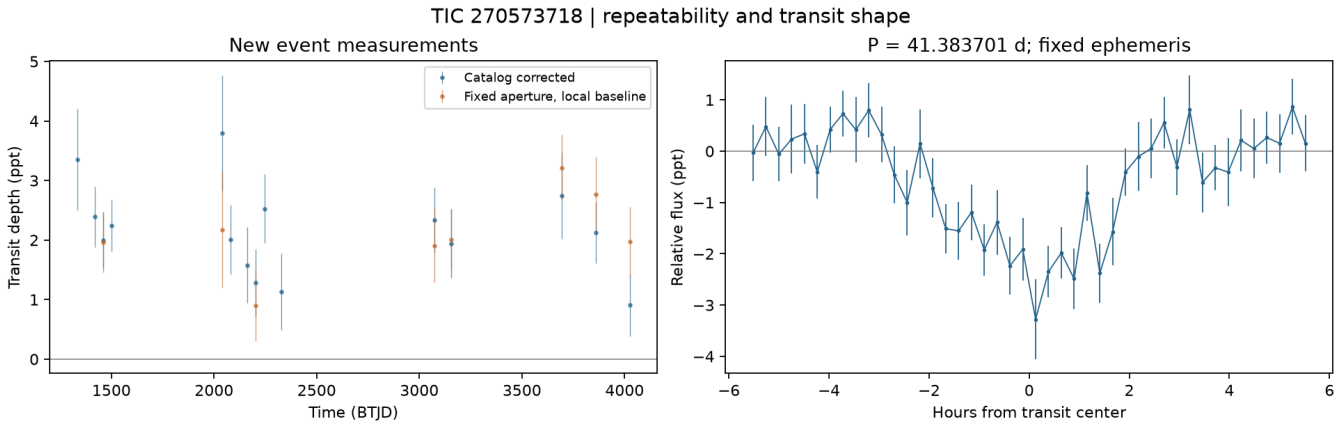
## Strong long-period planet candidate; target favored with a sector caveat

The 41.383701-day signal remains a strong long-period planet candidate. The available data favor the target over the bright nearby sources. A missing probability calculation and one baseline-sensitive recent event prevent a stronger claim.

| Parameter | Value | Evidence |
|---|---|---|
| Period / epoch | 41.38370104 d<br>BTJD 1334.903753 | Existing batch physical fit; BTJD = BJD(TDB) - 2457000 |
| Duration / nominal radius | 3.763 h<br>2.12 +/- 0.22 Earth radii | Conditional on the nominated host |
| New corrected / aperture amplitude | 2.033 +/- 0.165 ppt<br>2.111 +/- 0.244 ppt | 15 / 8 adequately covered events; all signs retained |
| New pixel sectors | 65, 88, 94, 98 | 4 fresh 15 x 15-pixel TESS cutouts |

The batch retains 13 events in 13 sectors. All 15 adequately covered events in the new corrected-light-curve measurement are positive, giving 2.033 +/- 0.165 ppt; eight saved fixed-aperture events give 2.111 +/- 0.244 ppt. The adopted 3.76-hour duration and 2.13 +/- 0.22 Earth-radius model are plausible. However, the unconstrained transit fit has a strong impact-parameter/density degeneracy: its density ratio is about 0.057, while the catalog-density solution has a small fit penalty. The adopted density is a prior, not independent confirmation.

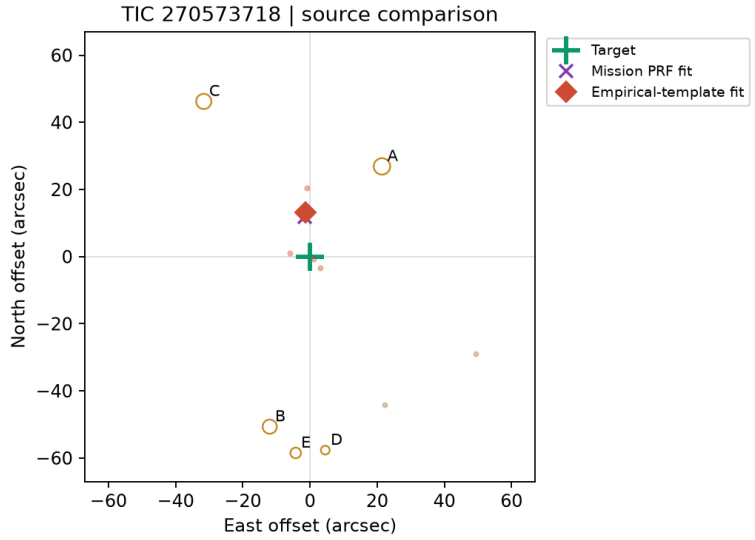
The sector-98 event falls to about 0.95 +/- 0.59 ppt with the quadratic baseline and is not independently decisive. A phase-0.5 dip appears at 3.5 formal sigma in the reused corrected light curve, but is only 0.10 +/- 0.25 ppt in the independently rebaselined fixed aperture. It is not established as a secondary eclipse. The original FPP run failed during cutout retrieval; FPP and NFPP are missing, not zero. No replacement FPP calculation is claimed in this report.




**Next decisive step.** Obtain another source-resolved primary transit and monitor the bright 34.4-arcsec neighbour. Audit the sector-98 event and the predicted half-phase window. Then rerun a complete, single-cadence FPP calculation with the scene fixed and every background scenario present.

# TIC 270573718: where is the dimming?

Sectors 65, 88 and 94 favor the target. A linear-baseline sector-98 map initially favors the bright neighbour Gaia DR3 4792202914059873280, about 34.4 arcsec away, but that preference reverses with a quadratic local baseline. The joint four-sector comparison favors the target over this neighbour by about 34-35 diagnostic loss units. The unrestricted fit is still offset by roughly 12-13 arcsec, so the conclusion is a target preference, not a precise clean localization. Leaving sector 98 out moves the empirical joint fit to just 1.6 arcsec from the target, showing that this recent event drives the larger offset.




**Nearby alternatives**

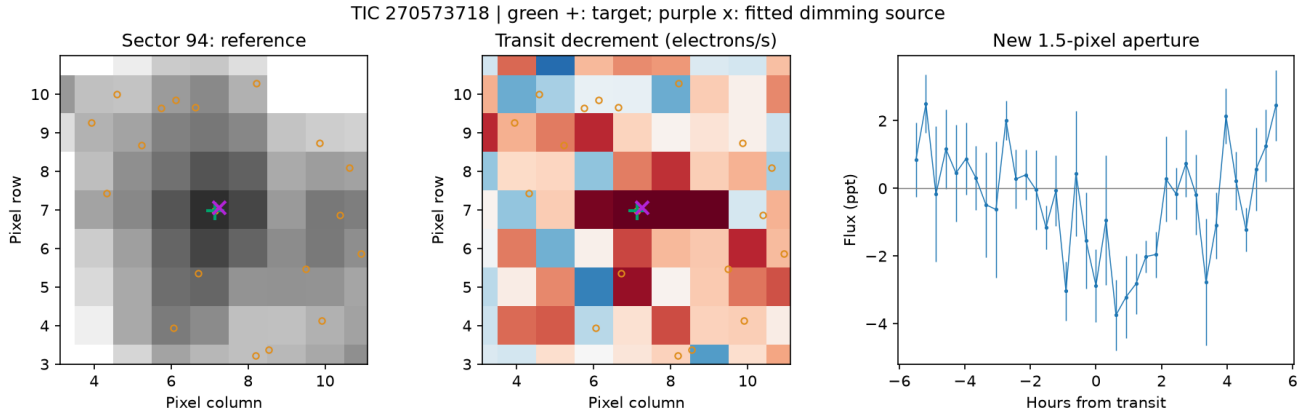
**A**: 34.4 arcsec; eclipse ~1%
Gaia DR3 4792202914059873280

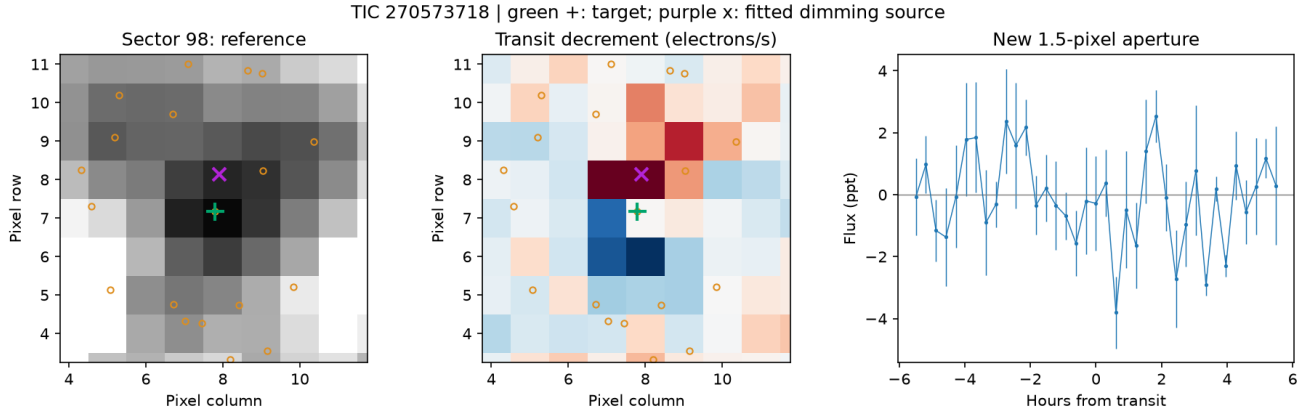
**B**: 52.1 arcsec; eclipse ~3%
Gaia DR3 4792202776620920448

**C**: 56.0 arcsec; eclipse ~2%
Gaia DR3 4792202948419610752

Labels correspond to the sky chart. The full source list is retained in the evidence JSON.

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The illustrated image fits use the linear baseline. Alternative quadratic-baseline results are retained in the data package and discussed where they change the interpretation. No position in this chart is a calibrated exclusion boundary.

# TIC 116046773

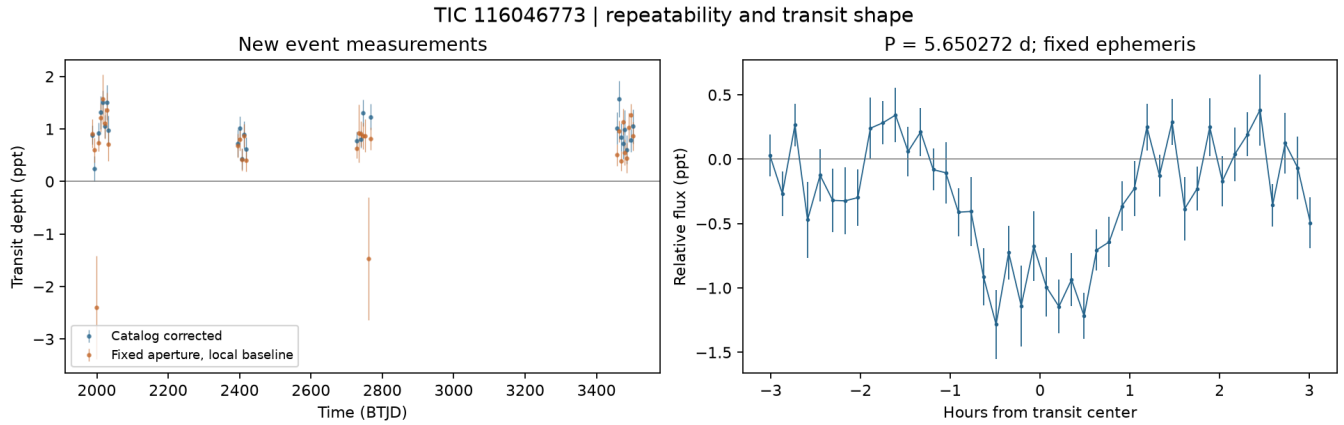
## Real transit-like signal; planet versus blended eclipse unresolved

The 5.650272-day dip is convincing, but I cannot reliably assign it to the nominated star. A planet on the target is plausible; a planet or eclipsing binary on a close companion remains a serious alternative. This is the weakest host assignment of the seven.

| Parameter | Value | Evidence |
|---|---|---|
| Period / epoch | 5.65027186 d<br>BTJD 1987.882622 | Existing batch physical fit; BTJD = BJD(TDB) - 2457000 |
| Duration / nominal radius | 2.054 h<br>2.29 +/- 0.24 Earth radii | Conditional on the nominated host |
| New corrected / aperture amplitude | 0.902 +/- 0.062 ppt<br>0.769 +/- 0.059 ppt | 25 / 30 adequately covered events; all signs retained |
| New pixel sectors | 25, 52, 79, 80 | 4 fresh 15 x 15-pixel TESS cutouts |

The new corrected and fixed-aperture amplitudes are 0.902 +/- 0.062 and 0.769 +/- 0.059 ppt. All 25 corrected events and 28 of 30 aperture events have positive depths. The apparent fixed-aperture brightenings in sectors 25, 52 and 53 become 0.688 +/- 0.226, 0.736 +/- 0.138 and 0.809 +/- 0.162 ppt dimmings under local baselines. No convincing phase-0.5 secondary or odd/even mismatch is found. This rehabilitates the signal, but does not identify its source.

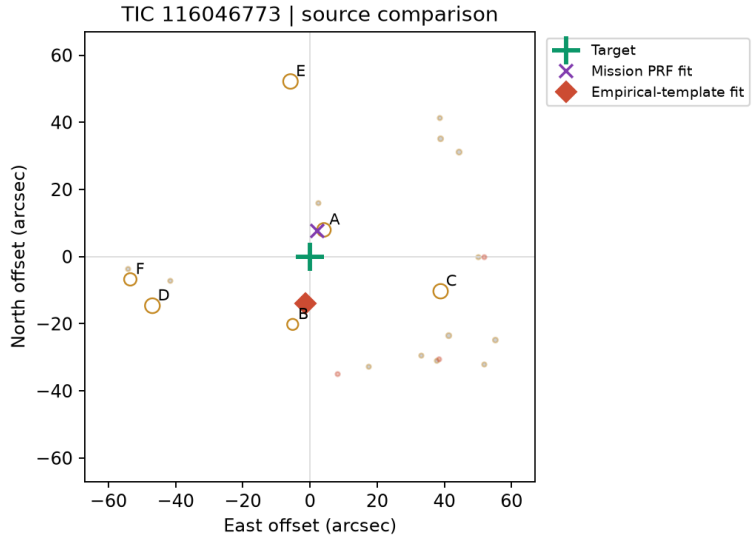
Fresh Gaia data put the close star 8.97 arcsec from the target, with parallax 15.9885 +/- 0.0656 mas versus 15.9929 +/- 0.0110 mas and similar proper motions. It is very likely a wide physical companion, at a projected separation near 560 AU. That does not make the 5.65-day signal an eclipse of the wide pair; the faint companion could itself host a close eclipsing system. Its RP flux ratio is about 0.00744, so a roughly 9-12% eclipse could reproduce the observed dip before throughput corrections. The nominal 2.29-Earth-radius target fit cannot be applied to that companion. The saved 12.18% FPP is almost entirely an assumed unresolved-companion planet scenario and does not resolve this measured scene ambiguity.




**Next decisive step.** Highest priority for source identification: obtain resolved time-series photometry of both stars during the same event. Nine arcseconds is readily separable in suitable ground-based imaging, and a ~10% companion eclipse would be much easier to detect than the ~0.08% target dip. Do not promote this to an on-target planet without that check.

# TIC 116046773: where is the dimming?

The calibrated mission PRF favors the direction of Gaia DR3 4608570247290179328 in three sectors, while sector 80 favors the target. The joint companion preference is only about 4.5 loss units and changes sign with the empirical template. Changing to quadratic baselines also produces inconsistent best-fitting sources, including stars too faint to explain the dip. Reference-image PRF residuals are 4.5-5.7% of peak, much larger than in the other fields. These facts indicate unreliable source discrimination rather than a secure detection on any particular neighbour.




**Nearby alternatives**

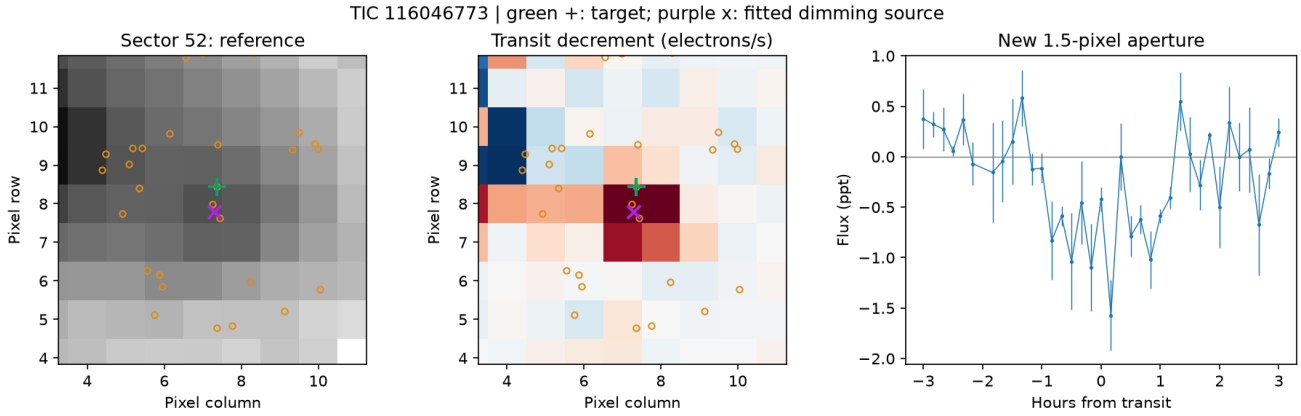
**A**: 9.0 arcsec; eclipse ~9%
Gaia DR3 4608570247290179328

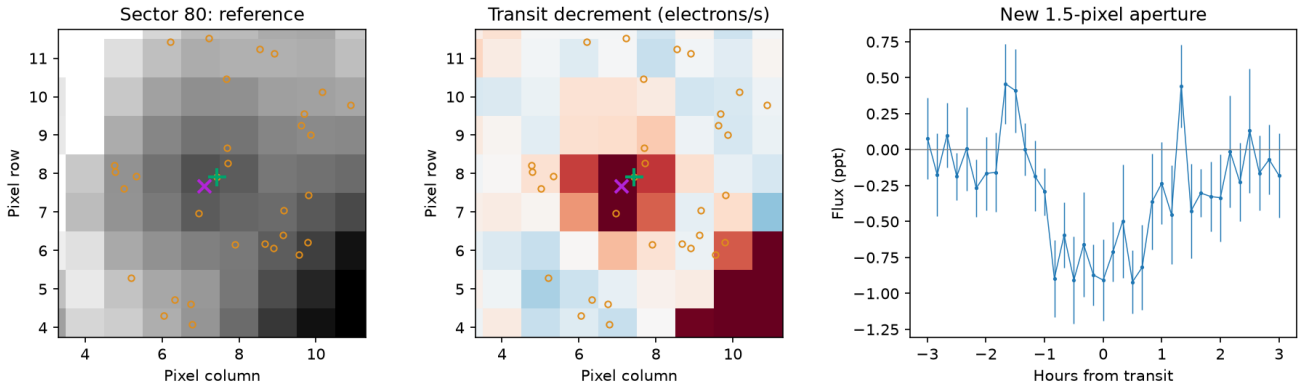
**B**: 20.9 arcsec; eclipse ~88%
Gaia DR3 4608570178569093120

**C**: 40.2 arcsec; eclipse ~12%
Gaia DR3 4608570178570702080

Labels correspond to the sky chart. The full source list is retained in the evidence JSON.

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The illustrated image fits use the linear baseline. Alternative quadratic-baseline results are retained in the data package and discussed where they change the interpretation. No position in this chart is a calibrated exclusion boundary.

# TIC 382098683

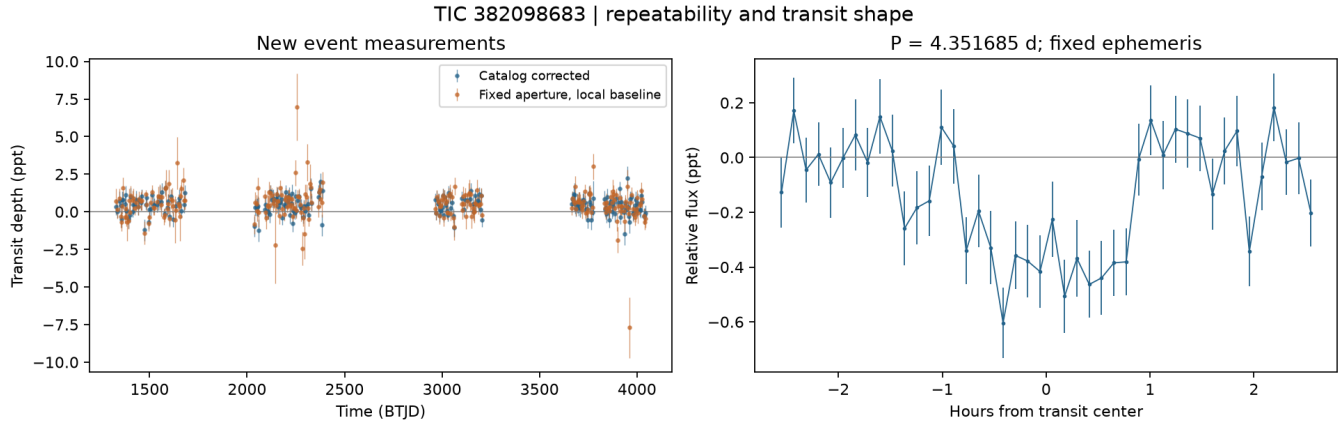
## Strong small-planet candidate; modest target preference

The 4.351685-day signal is a credible small-planet candidate. Its recurrence across many observing sectors is stronger evidence than any one pixel map. The target is the best combined catalogued source, but the available localization does not yet securely exclude nearby hosts.

| Parameter | Value | Evidence |
|---|---|---|
| Period / epoch | 4.35168457 d<br>BTJD 1329.139016 | Existing batch physical fit; BTJD = BJD(TDB) - 2457000 |
| Duration / nominal radius | 1.738 h<br>1.31 +/- 0.14 Earth radii | Conditional on the nominated host |
| New corrected / aperture amplitude | 0.437 +/- 0.037 ppt<br>0.500 +/- 0.041 ppt | 209 / 240 adequately covered events; all signs retained |
| New pixel sectors | 61, 87, 96, 97 | 4 fresh 15 x 15-pixel TESS cutouts |

The batch retains 184 events across 42 sectors. New measurements give 0.437 +/- 0.037 ppt in corrected photometry and 0.500 +/- 0.041 ppt in the saved aperture. The local-baseline analysis removes the extreme negative sector measurements; sector 97 changes from a nominal -8.03 ppt brightening to 0.23 +/- 0.24 ppt, a low-information result. Odd/even depths are compatible, and there is no positive phase-0.5 secondary. The nominal target-host radius is 1.31 +/- 0.14 Earth radii.

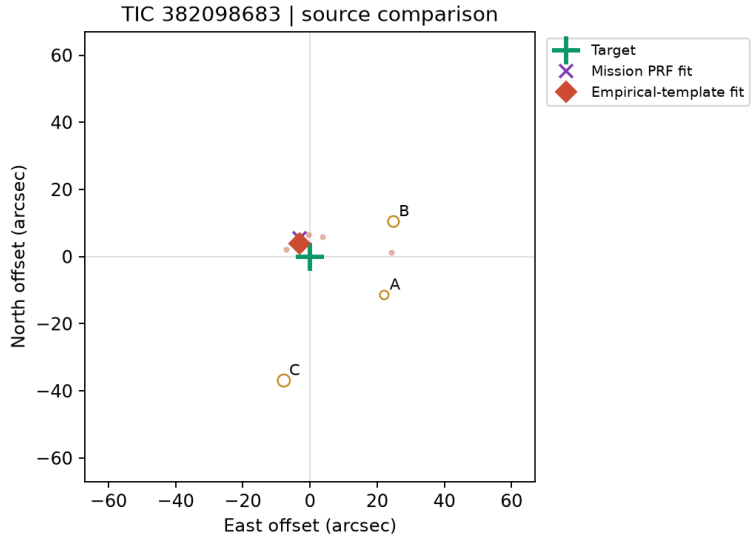
Three neighbours within 38 arcsec can plausibly mimic the dip by eclipse in the rough RP-band calculation. The 27.0-arcsec star would need only about an 8% eclipse before aperture losses. The saved FPP is 14.94%, but 14.82 percentage points are the planet-on-an-unresolved-companion scenario; the sum of stellar-eclipse scenarios is only about 0.10% within that conditional model. This is predominantly uncertainty about the host and radius, not strong model evidence for a stellar eclipse. Instrumental false alarms are outside that probability model.




**Next decisive step.** Prioritize resolved photometry of the 24.9-, 27.0- and 37.7-arcsec neighbours, followed by high-resolution imaging for closer companions. A source fit using all 42 pixel sectors should improve the weak four-sector localization.

# TIC 382098683: where is the dimming?

New sectors 61 and 87 prefer the target, sector 96 is weak, and sector 97 weakly prefers Gaia DR3 4762638489497421184. The combined result prefers the target over that 27.0-arcsec neighbour by only about 6-7 diagnostic loss units. The fitted position is 5-6 arcsec from the target. Quadratic baselines preserve the broad pattern: most sectors favor the target, while the low-significance sector-97 event stack remains ambiguous. This is useful support, but not a firm exclusion of the neighbour.




**Nearby alternatives**

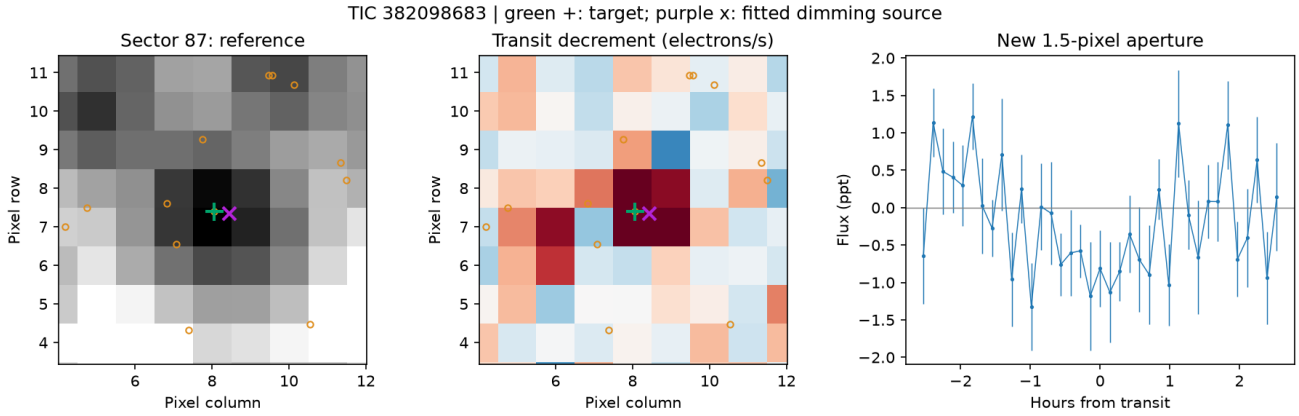
**A**: 24.9 arcsec; eclipse ~66%
Gaia DR3 4762638489497588352

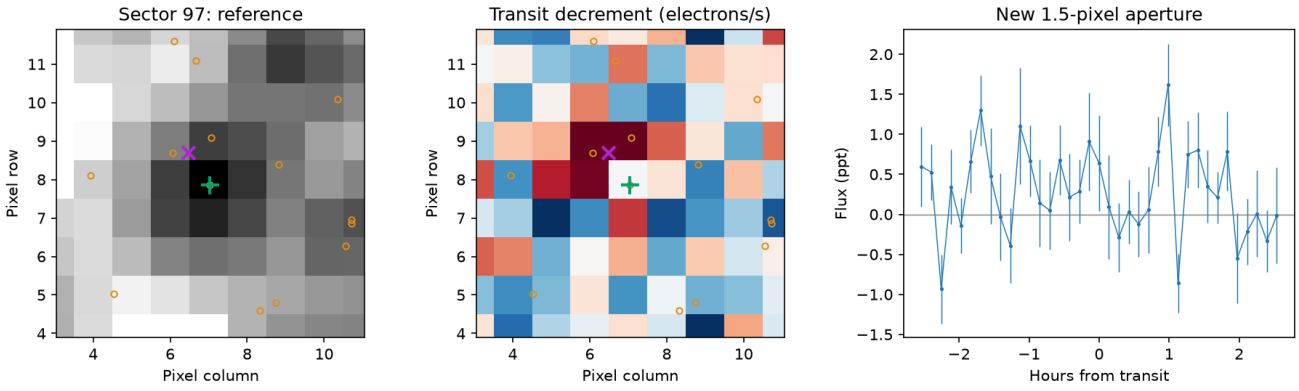
**B**: 27.0 arcsec; eclipse ~8%
Gaia DR3 4762638489497421184

**C**: 37.7 arcsec; eclipse ~6%
Gaia DR3 4762639211051928064

Labels correspond to the sky chart. The full source list is retained in the evidence JSON.

* Gaia-epoch separation, not rounded separation from the older TIC coordinate. ** Mean batch dip, Gaia RP flux ratio and equal target/neighbour throughput; these are illustrative estimates, not strict bounds or exact TESS-band predictions. Faint orange dots are less capable sources; circles are approximately capable. Pale red points show separate-sector empirical source fits. Crosses/diamonds show joint fits.




The illustrated image fits use the linear baseline. Alternative quadratic-baseline results are retained in the data package and discussed where they change the interpretation. No position in this chart is a calibrated exclusion boundary.

# TIC 262843927

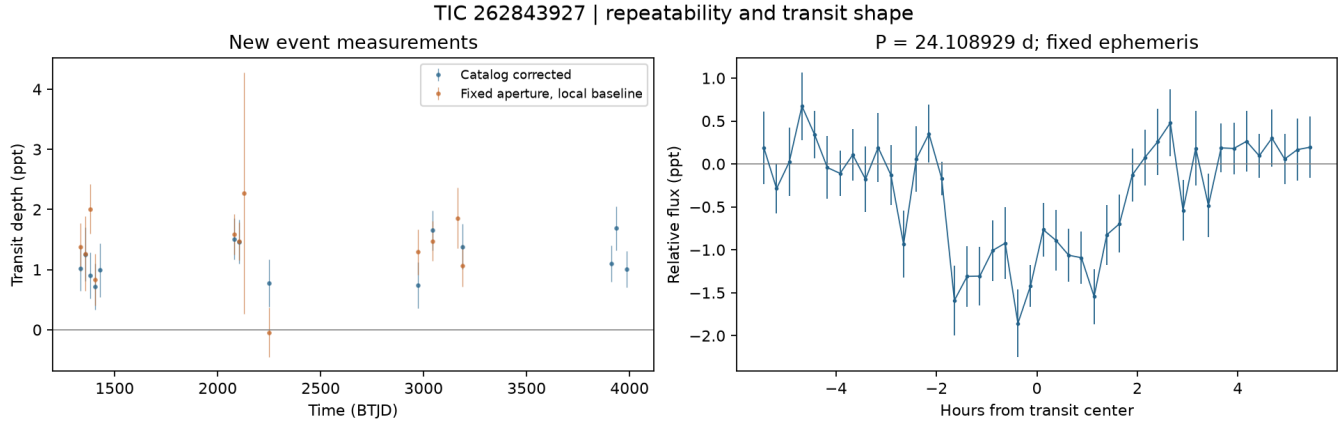
## Strong planet candidate; target strongly favored

This candidate improves substantially after the deeper checks. The 24.108929-day dip is persistent, new sector-96 pixels recover it, and the source fits favor the target. The previously alarming off-phase dip is not a stable secondary eclipse under event-by-event measurement.

| Parameter | Value | Evidence |
|---|---|---|
| Period / epoch | 24.10892866 d<br>BTJD 1332.787714 | Existing batch physical fit; BTJD = BJD(TDB) - 2457000 |
| Duration / nominal radius | 3.710 h<br>1.99 formal error unavailable Earth radii | Conditional on the nominated host |
| New corrected / aperture amplitude | 1.177 +/- 0.097 ppt<br>1.303 +/- 0.149 ppt | 14 / 12 adequately covered events; all signs retained |
| New pixel sectors | 64, 69, 96 | 3 fresh 15 x 15-pixel TESS cutouts |

All 14 adequately sampled corrected events are positive, with amplitude 1.177 +/- 0.097 ppt; the saved aperture gives 1.303 +/- 0.149 ppt. The primary has no significant odd/even difference and no phase-0.5 secondary. The earlier pipeline's phase-0.3876 dip, formally 11.8 sigma in its fold, falls to 0.069 +/- 0.216 ppt in corrected data and 0.093 +/- 0.199 ppt in the aperture after local fitting and event-scatter inflation. Its individual measurements are strongly inconsistent with a constant recurring depth.

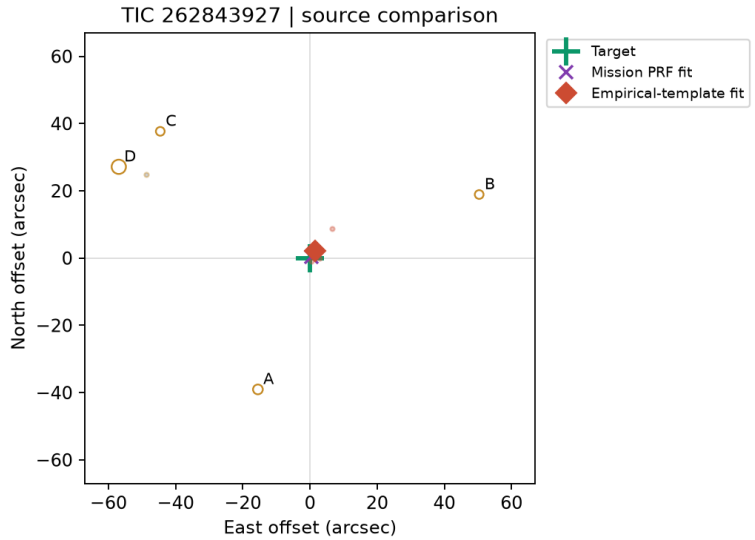
A smaller residual near phase 0.086 remains at about 3.8 formal sigma in corrected flux and 3.1 sigma in the saved aperture. It was selected from a phase search and is not a secure independent signal or secondary. The approximately 1.99-Earth-radius fit lands at impact parameter zero, where the original covariance calculation is singular; its missing radius error is a fitting limitation, not evidence for a binary. The saved FPP is 4.35%, almost entirely an unresolved-companion planet alternative. NFPP reported as zero is a numerical model result, not proof that all contaminating sources are impossible.




**Next decisive step.** Move this into the highest-priority follow-up group with TIC 193718989. Observe one resolved primary, obtain high-resolution imaging, and refit the central-transit geometry using a boundary-aware posterior or profile likelihood. Check the phase-0.086 residual without treating its trial-selected significance as validation.

# TIC 262843927: where is the dimming?

New sectors 64, 69 and 96 all prefer the target among catalogued positions. The joint source is 0.5 arcsec from the target with the mission PRF and 2.6 arcsec with the empirical template. The best competing catalogued position is worse by roughly 38 diagnostic loss units in either treatment. The closest Gaia neighbour is about 42 arcsec away, giving this case a cleaner resolved scene than most of the batch.




**Nearby alternatives**

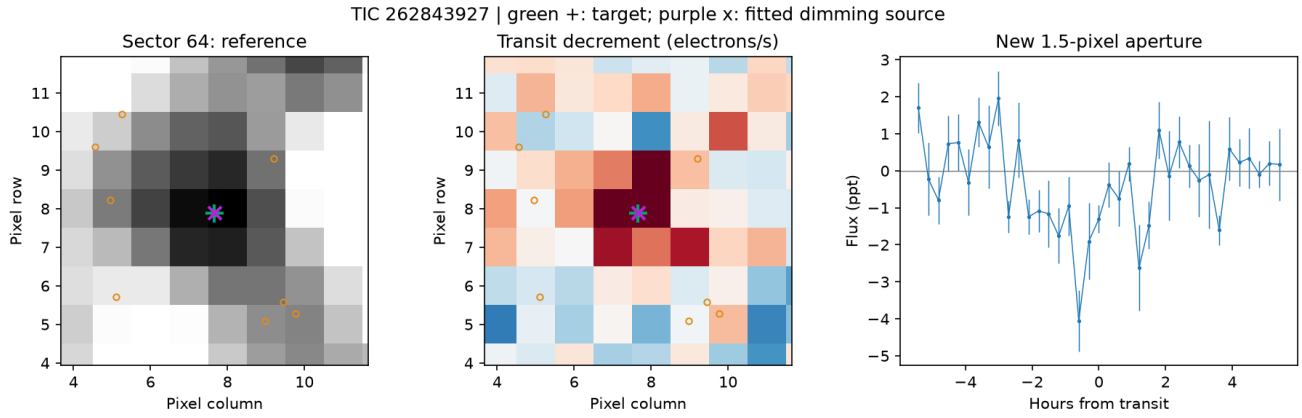
**A**: 42.0 arcsec; eclipse ~48%
Gaia DR3 4673750838475299968

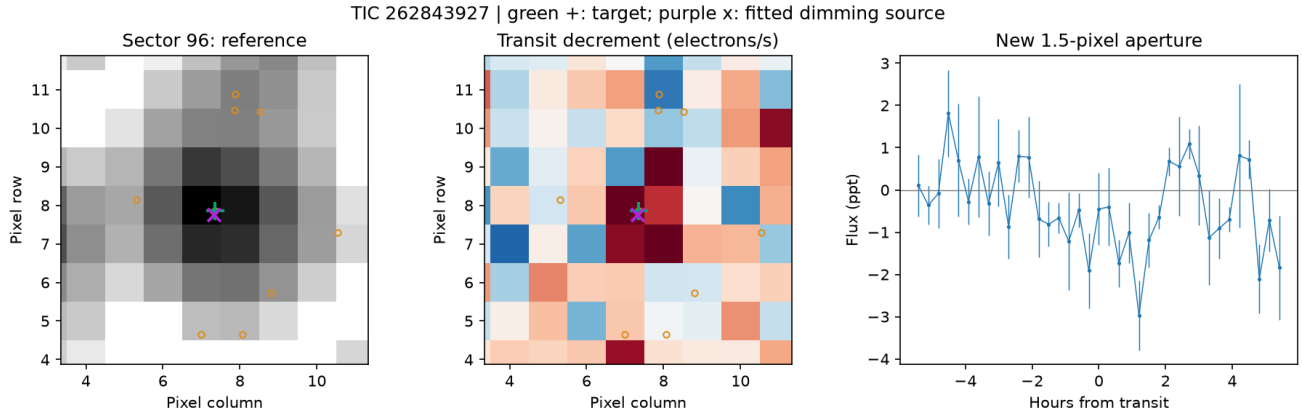
**B**: 53.8 arcsec; eclipse ~87%
Gaia DR3 4673843957661288960

**C**: 58.4 arcsec; eclipse ~80%
Gaia DR3 4673750945850295936

Labels correspond to the sky chart. The full source list is retained in the evidence JSON.

* Gaia-epoch separation, not rounded separation from the older TIC coordinate. ** Mean batch dip, Gaia RP flux ratio and equal target/neighbour throughput; these are illustrative estimates, not strict bounds or exact TESS-band predictions. Faint orange dots are less capable sources; circles are approximately capable. Pale red points show separate-sector empirical source fits. Crosses/diamonds show joint fits.




The illustrated image fits use the linear baseline. Alternative quadratic-baseline results are retained in the data package and discussed where they change the interpretation. No position in this chart is a calibrated exclusion boundary.

# TIC 201247574

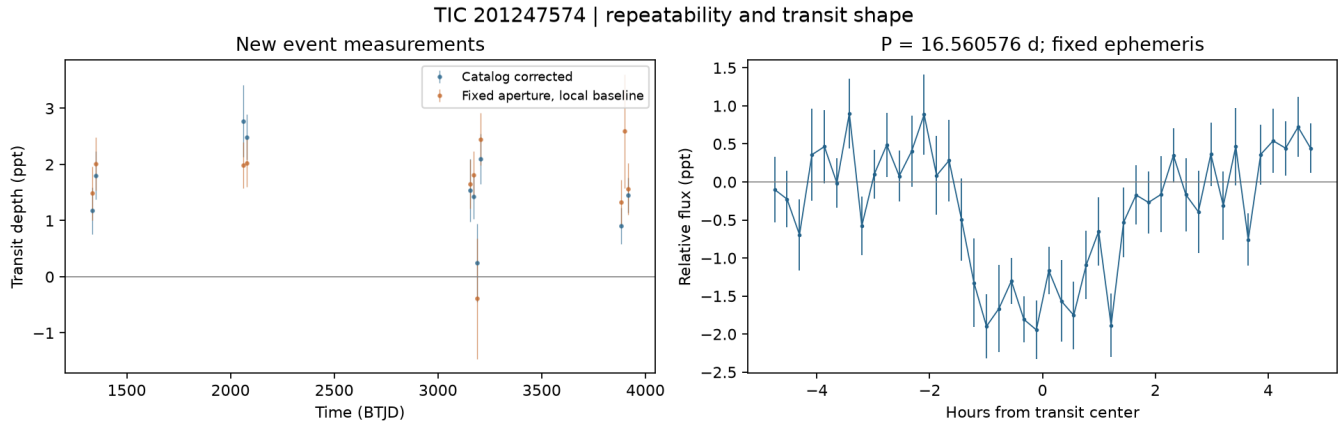
## Planet candidate; target favored and aperture concern reduced

The 16.560576-day signal has a stronger planet case than the original aperture inconsistency suggested. The target is favored by all three new pixel sectors, and the missing sector-96 dip reappears after local baseline treatment. Some temporal depth variation remains in the corrected product.

| Parameter | Value | Evidence |
|---|---|---|
| Period / epoch | 16.56057647 d<br>BTJD 1333.609906 | Existing batch physical fit; BTJD = BJD(TDB) - 2457000 |
| Duration / nominal radius | 3.240 h<br>2.44 +/- 0.27 Earth radii | Conditional on the nominated host |
| New corrected / aperture amplitude | 1.554 +/- 0.196 ppt<br>1.780 +/- 0.143 ppt | 10 / 11 adequately covered events; all signs retained |
| New pixel sectors | 68, 95, 96 | 3 fresh 15 x 15-pixel TESS cutouts |

Ten of ten adequately covered corrected events are positive. The new corrected amplitude is 1.554 +/- 0.196 ppt, including inflation for event scatter; the saved fixed aperture gives 1.780 +/- 0.143 ppt. Corrected-event depths show moderate heterogeneity (constant-depth p about 0.022), whereas aperture depths are compatible with a constant value (p about 0.48). Sector 95 becomes 1.50 +/- 0.44 ppt and sector 96 becomes 1.57 +/- 0.46 ppt, resolving their apparent deep-versus-absent conflict. A 2-sigma odd/even difference is not enough to establish a binary.

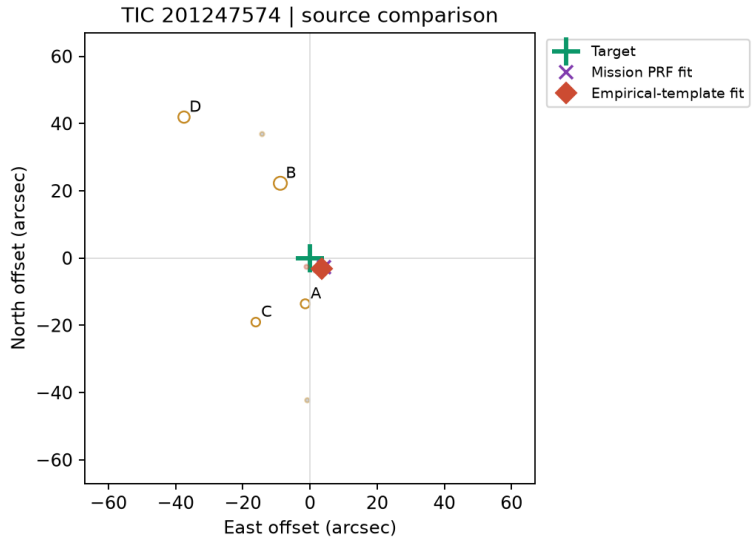
The half-phase secondary is not detected in the aperture, and the earlier phase-0.4324 feature falls to about 0.11 +/- 0.36 ppt. The Gaia/VSX solar-like variability classification is consistent with baseline changes, not a binary verdict. The batch's 2.44 +/- 0.27 Earth-radius estimate is plausible if the target is the host. Saved FPP/NFPP are 4.18%/0.367%; most of the remaining FPP is the unresolved-companion planet scenario. The 13.7-arcsec neighbour would require an approximately 84% eclipse in the two-star equal-throughput estimate, while a brighter star 23.9 arcsec away needs only about 9%.




**Next decisive step.** Upgrade its priority relative to the original HOLD. Obtain resolved primary-transit photometry, especially of the 13.7- and 23.9-arcsec neighbours, and compare independent aperture depths across epochs before interpreting the corrected-flux depth variation as astrophysical.

# TIC 201247574: where is the dimming?

New sectors 68, 95 and 96 all favor the target. Joint fits are about 4.6-4.9 arcsec from the target; the closest alternate position, Gaia DR3 4918283667602203520 at 13.7 arcsec, is worse by about 22 diagnostic loss units. The offset is not literally zero, and model errors are not calibrated, so the appropriate conclusion is good relative target support rather than a precise validation radius.




**Nearby alternatives**

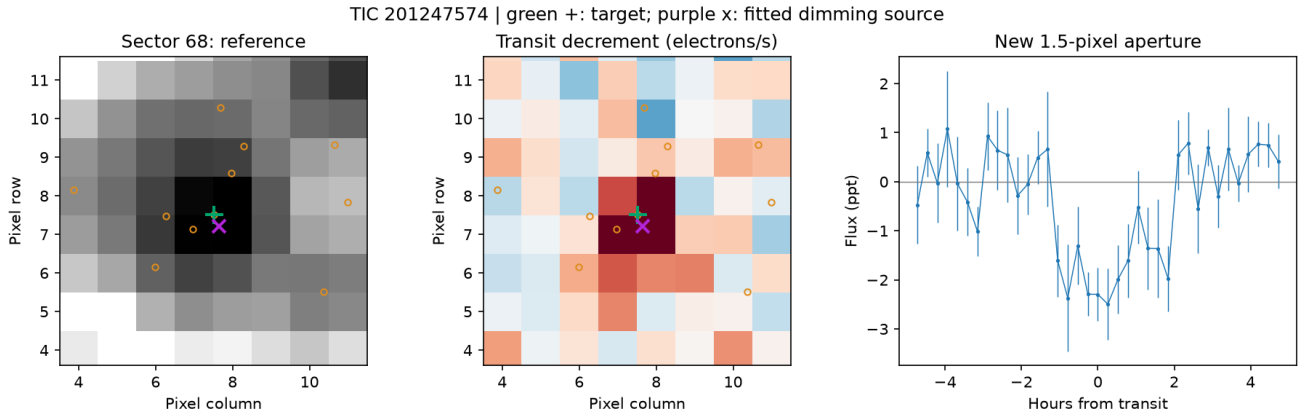
**A**: 13.7 arcsec; eclipse ~84%
Gaia DR3 4918283667602203520

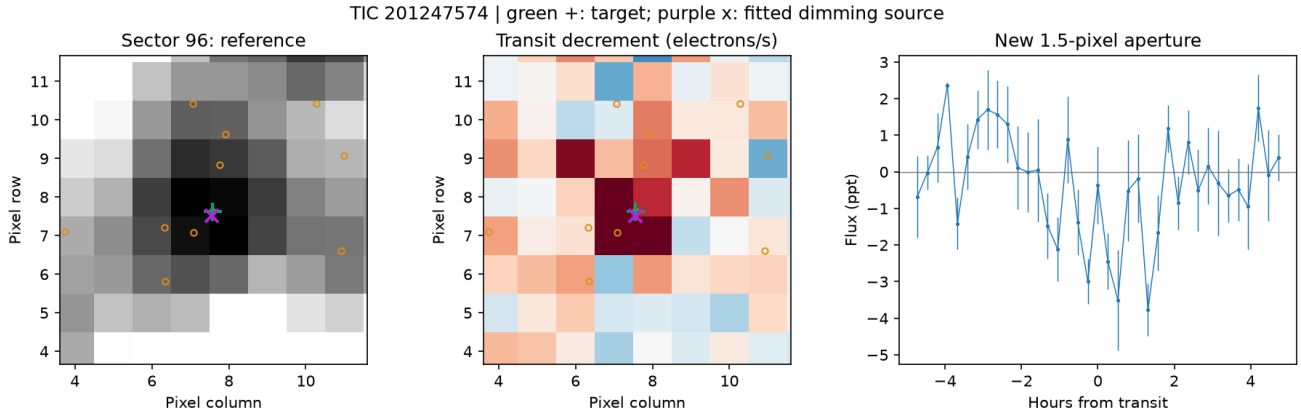
**B**: 23.9 arcsec; eclipse ~9%
Gaia DR3 4918284427812054272

**C**: 25.0 arcsec; eclipse ~35%
Gaia DR3 4918283671898155264

Labels correspond to the sky chart. The full source list is retained in the evidence JSON.

* Gaia-epoch separation, not rounded separation from the older TIC coordinate. ** Mean batch dip, Gaia RP flux ratio and equal target/neighbour throughput; these are illustrative estimates, not strict bounds or exact TESS-band predictions. Faint orange dots are less capable sources; circles are approximately capable. Pale red points show separate-sector empirical source fits. Crosses/diamonds show joint fits.




The illustrated image fits use the linear baseline. Alternative quadratic-baseline results are retained in the data package and discussed where they change the interpretation. No position in this chart is a calibrated exclusion boundary.

# Methods, limits and follow-up priorities

## Photometry

The fitted period, epoch and duration were fixed to the batch physical fit. Each adequately sampled event was fit with an exposure-averaged trapezoid plus a separate local linear baseline extending to three transit durations on each side (capped below half an orbit). The ingress/egress scale was 15% of duration. At least three in-event samples and four baseline samples on each side were required. Event signs were not used to select data. Errors were inflated using time-binned residuals and excess event-to-event scatter. Counts differ from the original pipeline because this is a fresh coverage-based measurement, not a replay of its retained-event mask.

## Pixels and localization

Twenty-five 15 x 15-pixel cutouts were retrieved from MAST. Cadences with the standard 17087 quality mask or nonfinite image values were removed. A frame-by-frame sky plane was estimated from low-flux outer pixels, avoiding bright catalogued sources. Local transit-template regressions produced per-pixel decrement and uncertainty maps. Gaia coordinates were propagated from epoch 2016 using proper motions. Public TESS PRFs were interpolated at the camera/CCD location; an out-of-transit scene fit allowed a common image-position correction. Each candidate-source fit included free positive signal amplitude and a background plane. A second fit used an empirical target-image template. [1,2]

Joint image scores are divided by each sector's best-fit reduced chi-square when it exceeds one. They remain diagnostic: spatial pixel covariance, calibration mismatch, source-model incompleteness and timing uncertainty are not fully propagated. The report does not turn loss differences into host percentages or quote local Hessian ellipses as trustworthy exclusion radii. Leave-one-sector-out fits test dependence on individual observations. Four difficult cases also received quadratic-baseline and retained-event sensitivity checks. Only the selected sectors, not every available raw pixel sector, were re-extracted.

## Practical observing order

**Best planet follow-up:** TIC 193718989 and TIC 262843927, then TIC 201247574 and TIC 270573718. The first pair combines strong repetition with cleaner localization. The long period of TIC 270573718 makes a missed event costly, so an accurately scheduled observation may take operational priority.

**Most decisive contamination checks:** resolve TIC 116046773 and its 9-arcsec companion; monitor the faint neighbours of TIC 188830138 and TIC 382098683. Rejecting deep eclipses on the contaminants is often easier than detecting a few-hundred-ppm transit on the target. High-resolution imaging then addresses unresolved companions; spectroscopy/RV can test stellar-mass alternatives.

No new radial velocities, high-resolution imaging, ground-based transits, complete global false-alarm experiment, or fresh seven-target FPP campaign was obtained. These limits prevent confirmation or validation. Predictions should propagate period/epoch covariance and any transit-timing uncertainty before an observing run. No observatory-specific visibility or future transit schedule is claimed.

# Sources and reproducibility

## Primary data and methods

**[1] MAST TESS archive and public TESScut service**
https://archive.stsci.edu/missions-and-data/tess
25 public sector cutouts downloaded in this task; download_manifest.json records URLs and sizes. Per-sector JSON stores SHA-256 hashes.

**[2] TESS science data products and PRF models**
https://archive.stsci.edu/missions-and-data/tess/data-products
Difference-image/PRF source methodology; public start_s0004 PRF grid. Calibration inputs are listed in every pixel-result file.

**[3] Giacalone et al., TRICERATOPS validation and vetting**
https://arxiv.org/abs/2002.00691
Definitions and interpretation of FPP/NFPP and astrophysical scenarios. Existing batch probabilities are reported separately from this review's measurements.

**[4] Public TIC 193718989 candidate analysis**
https://github.com/eumemic/exoplanets/tree/main/candidates/TIC193718989_P24.533
Retrieved vetting.json, transit_fit.json and triceratops_gaia.json. Same period and consistent radius; shared underlying TESS observations.

**[5] Gaia DR3 archive**
https://gea.esac.esa.int/archive/
Fresh 90-arcsec field queries around the seven TIC positions. Exact ADQL request and CSV response retained, with source IDs preserved as strings for analysis.

**[6] ExoFOP TESS candidate lists**
https://exofop.ipac.caltech.edu/tess/
Fresh TOI and CTOI tables retrieved 18 October 2026. No exact TIC-ID match for these seven in either downloaded table. Absence is not proof of novelty; TIC 193718989 already has a public candidate report.

## Local evidence used

The source is the latest batch-one packet set at `D:/tesshunter_s1/packets/s1_b1`, with funnel light curves and saved aperture arrays at `D:/tesshunter_s1/funnel/s1_b1`. The batch result was generated at 2026-10-18 14:21 UTC. The original survey and its verdicts were not modified. Existing broad batch-review files were used to locate the inputs; the new numerical analysis was performed by the scripts supplied with this report.

## How to inspect or repeat the analysis

The evidence package includes the Python scripts, extracted batch inputs, fresh catalog results, numerical event/source measurements, image maps and plots. Run inspect_inputs.py only if you need to regenerate the initial input snapshot from the local batch. fetch_public.py and fetch_gaia.py retrieve public data; analyze_photometry.py, analyze_pixels.py, refine_localization.py and baseline_sensitivity.py perform the measurements. audit_catalogs.py builds the contamination table; make_summary_figures.py and build_report.py build this report.

The large raw FITS cutouts remain in the task's analysis/cutouts folder and are omitted from the compact evidence ZIP. Original full light-curve arrays remain in the survey cache. The scripts use those local paths, so the ZIP alone is an evidence record rather than a fully self-contained raw-data distribution. All computations, source positions and limitations used in the conclusions are present in the JSON files.

*Interpretation date: 18 October 2026. Classification terms such as "planet favored" and "target favored" are qualitative assessments of the available evidence; they are not official catalog dispositions or calibrated probabilities.*